

CHAPTER 2.

EVALUATING ARGUMENTS

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One particularly relevant application of logic is assessing the relative strength of philosophical claims. While the topics covered by philosophers are fascinating, it is often difficult to determine which positions on these topics are the right ones. Many students are led to think that philosophy is just a matter of opinion. After all, who could claim to know the final answer to philosophical questions?

It's not likely that anyone will ever know the final answer to deep philosophical questions. Yet there are clearly better and worse answers; and philosophy can help us distinguish them. This chapter will give you some tools to begin to distinguish which positions on philosophical topics are well-founded and which are not. When a person makes a claim about a philosophical subject, you should ask, "What are the arguments to support that claim?" Once you have identified an argument, you can use these tools to assess whether it's a good or bad one, whether the evidence and reasoning really support the claim or not.

In broad terms, there are two features of arguments that make them good: (1) the structure of the argument and (2) the truth of the evidence provided by the argument. Logic deals more directly with the structure of arguments. When we examine the logic of arguments, we are interested in whether the arguments have the right architecture, whether the evidence provided is the right sort of evidence to support the conclusion drawn. However, once we try to evaluate the truth of the conclusion, we need to know whether the evidence is true. We'll look at both of these considerations in what follows.

INFERENCE AND IMPLICATION: WHY CONCLUSIONS FOLLOW FROM PREMISES

An argument is a connected series of propositions, some of which are called premises and at least one of which is a conclusion. The premises provide the reasons or evidence that supports the conclusion. From the point of view of the reader, an argument is meant to persuade the reader that, once the premises are accepted as true, the conclusion follows from them. If the reader accepts the premises, then she ought to accept the conclusion. The act of reasoning that connects the premises to the conclusion is called an **inference**. A good argument supports a rational inference to the conclusion, a bad argument supports no rational inference to the conclusion.¹

1. This does not mean that bad arguments cannot be psychologically persuasive. In fact, people are often persuaded by bad arguments. However, a good philosophical assessment of an argument ought to rely purely on the rationality of its inferences.

Consider the following example:

1. All human beings are mortal.
2. Socrates is a human being.

3. / ∴ Socrates is mortal.

This argument asserts that Socrates is mortal. It does so by appealing to the fact that Socrates is a human being, together with the idea that all human beings are mortal. There is clearly a strong connection between the premises and conclusion. Imagine a reader who accepts both premises but denies the conclusion. This person would have to believe that Socrates is a human being and that all human beings are mortal, but still deny that Socrates is mortal. How could such a person maintain that belief? It just doesn't seem rational to believe the premises but deny the conclusion!

Now consider the following argument:

1. I saw a black cat today.
2. My knee is aching.

3. / ∴ It is going to rain.

Suppose that it does, in fact, rain and the person who advances this argument believes that it is going to rain. Is that person justified in their belief that it will rain? Not based on the argument presented here! In this argument, there is a very weak connection between the premises and the conclusion. So, even if the conclusion turns out to be true, there is no reason why a reader ought to accept the conclusion given these premises (there may be other reasons for thinking it is going to rain that are not provided here, of course). The point is that these premises do not provide the right sort of evidence to justify the conclusion.

So far, I have described the connection between premises and conclusion in terms of the psychological demand placed on a reader of the argument. However, we can describe this connection from another perspective. We can say that the premises of an argument **logically imply** a conclusion. Either way of speaking is correct. What they assert is that good arguments present a strong connection between the truth of the premises and the truth of the conclusion. In the next few sections, we will examine three different types of logical connection, each with its own rules for evaluation. Sometimes logical implication is guaranteed (as in the case of **deductive arguments**), sometimes the logical connection only ensures the conclusion is probable (as with **inductive** and **abductive arguments**).

DEDUCTIVE ARGUMENTS

Deductive arguments are the most common type of argument in philosophy, and for good reason. Deductive arguments attempt to demonstrate that the conclusion *follows necessarily* from the premises. As long as the premises of a *good* deductive argument are true, the conclusion is true as a matter of logic. This means that if I know the premises are true, I know with one-hundred percent certainty that the conclusion is also true! This may be hard to believe; after all, how can we be absolutely certain about anything? But notice what I am saying: I am not saying that we know the conclusion is true with one-hundred percent certainty. I am saying that we can

be one-hundred percent certain the conclusion is true, *on the condition that* the premises are true. If one of the premises is false, then the conclusion is not guaranteed.

Here are two examples of good deductive arguments. They are both valid *and* have true premises. A **valid argument** is an argument whose premises guarantee the truth of the conclusion. That is, if the premises are true, then it is impossible for the conclusion to be false. A valid deductive argument whose premises are all true is called a **sound argument**.

1. If it rained outside, then the streets will be wet.
2. It rained outside.

3. / ∴ The streets are wet.

1. Either the world ended on December 12, 2012 or it continues today.
2. The world did not end on December 12, 2012.

3. / ∴ The world continues today.

Hopefully, you can see that these arguments present a close connection between the premises and conclusion. It seems impossible to deny the conclusion while accepting that the premises are all true. This is what makes them valid deductive arguments. To show what happens when similar arguments employ false premises, consider the following examples:

1. If Russia wins the 2018 FIFA World Cup, then Russia is the reigning FIFA world champion [in 2019].
2. Russia won the 2018 FIFA World Cup.

3. / ∴ Russia is the reigning FIFA world champion [in 2019].

1. Either snow is cold or snow is dry.
2. Snow is not cold.

3. / ∴ Snow is dry.

You may recognize that these arguments have the same structure as the previous two arguments. That is, each expresses the same connection between the premises and conclusion, and they are all deductively valid. However, these latter two arguments have at least one false premise and this false premise is the reason why these otherwise valid arguments reach a false conclusion. In the case of these arguments, the structure is good, but the evidence is bad.

Deductive arguments are either valid or invalid because of the form or structure of the argument. They are sound or unsound based on the form, plus the content. You might become familiar with some of the common forms of

arguments (many of them have names) and once you do, you will be able to tell when a deductive argument is invalid.

Now let's look at some invalid deductive arguments. These are arguments that have the wrong structure or form. Perhaps you have heard a playful argument like the following:

1. Grass is green.
2. Money is green.
- _____
3. / ∴ Grass is money.

Here is another example of the same argument:

1. All tigers are felines.
2. All lions are felines.
- _____
3. / ∴ All tigers are lions.

These arguments are examples of the fallacy of the *undistributed middle term*. The name is not important, but you may recognize what is going on here. The two types of objects in each conclusion are each a member of some third type, but they are not members of each other. So, the premises are all true, but the conclusions are false. If you encounter an argument with this structure, you will know that it is invalid.

But what do you do if you cannot immediately recognize when an argument is invalid? Philosophers look for counterexamples. A **counterexample** is a scenario in which the premises of the argument are true while the conclusion is clearly false. This automatically shows that it is *possible* for the argument's premises to be true and the conclusion false. So, a counterexample demonstrates that the argument is invalid. After all, validity requires that if the premises are all true, the conclusion cannot possibly be false. Consider the following argument, which is an example of a fallacy called *affirming the consequent*:

1. If it rained outside, then the streets will be wet.
2. The streets are wet.
- _____
3. / ∴ It rained outside.

Can you imagine a scenario where the premises are true, but the conclusion is false?

What if a water main broke and flooded the streets? Then the streets would be wet, but it may not have rained. It would still remain true that *if* it had rained, the streets would be wet, but in this scenario even if it didn't rain, the streets would still be wet. So, the scenario where a water main breaks demonstrates this argument is invalid.

The counterexample method can also be applied to arguments where there is no clear scenario that makes the premises true and the conclusion false, but we will have to apply it a little differently. In these cases, we need to imagine another argument that has exactly the same structure as the argument in question but uses propositions that more easily produce a counterexample. Suppose I made the following argument:

1. Most people who live near the coast know how to swim.
 2. Mary lives near the coast.
-
3. / $\therefore$ Mary knows how to swim.

I don't know if Mary knows how to swim, but I do know that this argument does not provide sufficient reasons for us to know that Mary knows how to swim. I can demonstrate this by imagining another argument with the same structure as this argument, but the premises of this argument are clearly true while its conclusion is false:

1. Most months in the calendar year have at least 30 days.
 2. February is a month in the calendar year.
-
3. / $\therefore$ February has at least 30 days.

To review, deductive arguments purport to lead to a conclusion that must be true if all the premises are true. But there are many ways a deductive argument can go wrong. In order to evaluate a deductive argument, we must answer the following questions:

- Are the premises true? If the premises are not true, then even if the argument is valid, the conclusion is not guaranteed to be true.
- Is the form of the argument a valid form? Does this argument have the exact same structure as one of the invalid arguments noted in this chapter or elsewhere in this book?²
- Can you come up with a counterexample for the argument? If you can imagine a case in which the premises are true but the conclusion is false, then you have demonstrated that the argument is invalid.

INDUCTIVE ARGUMENTS

Almost all of the formal logic taught to philosophy students is deductive. This is because we have a very well-established formal system, called first-order logic, that explains deductive validity.³ Conversely, most of the inferences we make on a daily basis are inductive or abductive. The problem is that the logic governing inductive and abductive inferences is significantly more complex and more difficult to formalize than deductive inferences.

The chief difference between deductive arguments and inductive or abductive arguments is that while the former arguments aim to guarantee the truth of the conclusion, the latter arguments only aim to ensure that the conclusion is *more probable*. Even the conclusions of the best inductive and abductive arguments may still turn out to be false. Consequently, we do not refer to these arguments as valid or invalid. Instead, arguments with good inductive and abductive inferences are **strong**; bad ones are **weak**. Similarly, strong inductive or abductive arguments with true premises are called **cogent**.

Here's a table to help you remember these distinctions:

2. Chapters 3 and 4 of this *Introduction* address types of fallacies. Fallacies are just systematic mistakes made within arguments. You can learn more examples of invalid argument forms in these chapters.
3. Chapter 3 introduces formal logic.

Terms used when evaluating several kinds of arguments

Quality of Inference	Deductive	Inductive	Abductive
Bad inference	Invalid	Weak	Weak
Good inference	Valid	Strong	Strong
Good inference + true premises	Sound	Cogent	Cogent

Inductive inferences typically involve an appeal to past experience in order to infer some further claim directly related to that experience. In its classic formulation, inductive inferences move from observed instances to unobserved instances, reasoning that what is not yet observed will resemble what has been observed before. Generalizations, statistical inferences, and forecasts about the future are all examples of inductive inference.⁴ A classic example is the following:

1. The Sun rose today.
 2. The Sun rose yesterday.
 3. The Sun has risen every day of human history.
-
4. / ∴ The Sun will rise tomorrow.

You might wonder why this conclusion is merely probable. Is there anything more certain than the fact that the Sun will rise tomorrow? Well, not much. But at some point in the future, the Sun, like all other stars, will die out and its light will become so faint that there will be no sunrise on the Earth. More radically, imagine an asteroid disrupting the Earth's rotation so that it fails to spin in coordination with our 24-hour clocks—in this case, the Sun would also fail to rise tomorrow. Finally, any inference about the future must always contain a degree of uncertainty because we cannot be certain that the future will resemble the past. So, even though the inference is very strong, it does not provide us with one-hundred percent certainty.

Consider the following, very similar inference, from the perspective of a chicken:

1. When the farmer came to the coop yesterday, he brought us food.
 2. When the farmer came to the coop the day before, he brought us food.
 3. Every day that I can remember, the farmer has come to the coop to bring us food.
-
4. / ∴ When the farmer comes today, he will bring food.

From a chicken's perspective, this inference looks equally as strong as the previous one. But this chicken will be surprised on that fateful day when the farmer comes to the coop with a hatchet to butcher her! From the chicken's perspective, the inference may appear strong, but from the farmer's perspective, it's fatally flawed. The chicken's inference shares some similarities with the following example:

4. You may notice that the inference from the previous section about Mary being able to swim could be rephrased as a kind of inductive argument. If it is true that most people who live near the coast can swim and Mary lives near the coast, then it follows that Mary probably can swim. This demonstrates an important difference between deductive and inductive arguments.

1. A recent poll of over 5,000 people in the USA found that 85% of them are members of the National Rifle Association.
 2. The poll found that 98% of respondents were strongly or very strongly opposed to any firearms regulation.
-
3. / $\therefore$ Support of gun rights is very strong in the USA.

While the conclusion of this argument may be true and certainly appears to be supported by the premises, there is a key weakness that undermines the argument. You may suspect that these polling numbers present unusually high support for guns, even in the USA.⁵ So, you may suspect that something is wrong with the data. But if I tell you that this poll was taken outside of a gun show, then you should realize that data may be correct, but the sample is clearly flawed. This reveals something important about inductive inferences. Inductive inferences depend on whether the sample set of experiences from which the conclusion is inferred are *representative* of the whole population described in the conclusion. In the cases of the chicken and gun rights, we are provided with a sample of experiences that are not representative of the populations in the conclusion. If we want to generalize about chicken farmer behavior, we need to sample the range of behaviors a farmer engages in. One chicken may not have enough data points to make a generalization about farmer behavior. Similarly, if we want to make a claim about the gun control preferences in the USA, we need to have a sample that represents all Americans, not just those who attend gun shows. The sample of experiences in a strong inductive argument must be representative of the conclusion that is drawn from it.

To review, strong inductive inferences lead to conclusions that are made more likely by the premises, but not guaranteed to be true. They are typically used to make generalizations, infer statistical probabilities, and make forecasts about the future. To evaluate an inductive inference, you should use the following guidelines:

- Are the premises true? Just like deductive arguments, inductive arguments require true premises to infer that the conclusion is likely to be true.
- Are the examples cited in the premises a large enough sample? The larger the sample, the greater the likelihood it is representative of the population as a whole, and thus the more likely inductive inferences made on the basis of it will be strong.

ABDUCTIVE ARGUMENTS

Abductive arguments produce conclusions that attempt to explain the phenomena found in the premises. From a commonsense point of view, we can think of abductive inferences as “reading between the lines,” “using context clues,” or “putting two and two together.” We typically use these phrases to describe an inference to an explanation that is not explicitly provided. This is why abductive arguments are often called an “inference to the best explanation.” From a scientific perspective, abduction is a critical part of hypothesis formation. Whereas the classic “scientific method” teaches that science is deductive and that the purpose of experimentation is to test a hypothesis (by confirming or disconfirming the hypothesis), it is not always clear how scientists arrive at a hypothesis. Abduction provides an explanation for how scientists generate likely hypotheses for experimental testing.

5. See, for instance, recent Gallup polling: 2019. “Guns.” <http://news.gallup.com/poll/1645/guns.aspx>.

Even though Sherlock Holmes is famous for declaring, in the course of his investigations, “Deduction, my dear Watson,” he probably should have said “Abduction”! Consider the following inference:

1. The victim’s body has multiple stab wounds on its right side.
 2. There was evidence of a struggle between the murderer and the victim.
-
3. / ∴ The murderer was left-handed.

You should recognize that the conclusion is not guaranteed by the premises, and so it is not a deductive argument. Additionally, the argument is not inductive, because the conclusion isn’t simply an extension from past experiences. This argument attempts to provide the best explanation for the evidence in the premises. In a struggle, two people are most likely to be standing face to face. Also, the killer probably attacked with his or her dominant hand. It would be unnatural for a right-handed person to stab with their left hand or to stab a person facing them on that person’s right side. So, the fact that the murderer is left-handed provides the most likely explanation for the stab wounds.

You use these sorts of inferences regularly. For instance, suppose that when you come home from work, you notice that the door to your apartment is unlocked and various items from the refrigerator are out on the counter. You might infer that your roommate is home. Of course, this explanation is not guaranteed to be true. For instance, you may have forgotten to lock the door and put away your food in your haste to get out the door. Abductive inferences attempt to reason to the most likely conclusion, not one that is guaranteed to be true.

What makes an abductive inference strong or weak? Good explanations ought to take account of all the available evidence. If the conclusion leaves some evidence unexplained, then it is probably not a strong argument. Additionally, extraordinary claims require extraordinary evidence. If an explanation requires belief in some entirely novel or supernatural entity, or generally requires us to revise deeply held beliefs, then we ought to demand that the evidence for this explanation is very solid. Finally, when assessing alternative explanations, we should heed the advice of “Ockham’s Razor.” William of Ockham argued that given any two explanations, the simpler one is more likely to be true. In other words, we should be skeptical of explanations that require complex mechanics, extensive caveats and exceptions, or an extremely precise set of circumstances, in order to be true.⁶

Consider the following arguments with identical premises:

1. There have been hundreds of stories about strange objects in the night sky.
 2. There is some video evidence of these strange objects.
 3. Some people have recalled encounters with extraterrestrial life forms.
 4. There are no peer-reviewed scientific accounts of extraterrestrial life forms visiting earth.
-
5. / ∴ There must be a vast conspiracy denying the existence of aliens.

6. While Ockham’s Razor is a good rule of thumb in evaluating explanations, there is considerable debate among philosophers of science about whether simplicity it is a feature of good scientific explanations or not.

1. There have been hundreds of stories about strange objects in the night sky.
 2. There is some video evidence of these strange objects.
 3. Some people have recalled encounters with extraterrestrial life forms.
 4. There are no peer-reviewed scientific accounts of extraterrestrial life forms visiting earth.
-
5. / ∴. The stories, videos, and recollections are probably the result of confusion, confabulation or exaggeration, or are outright falsifications.

Which is the more likely explanation?

To review, abductive inferences assert a conclusion that the premises do not guarantee, but which aims to provide the most likely explanation for the phenomena detailed in the premises. To assess the strength of an abductive inference, use the following guidelines:

- Is all the relevant evidence provided? If critical pieces of information are missing, then it may not be possible to know what the right explanation is.
- Does the conclusion explain all of the evidence provided? If the conclusion fails to account for some of the evidence, then it may not be the best explanation.
- Extraordinary claims require extraordinary evidence! If the conclusion asserts something novel, surprising, or contrary to standard explanations, then the evidence should be equally compelling.
- Use Ockham's Razor; recognize that the simpler of two explanations is likely the correct one.